SMARTBRIDGE
Let's Bridge the Gap

Project Initialization and Planning Phase

Date	21 October 2026
Team ID	EN25CS3010849
Project Title	Global Olympic Sports Performance Analysis
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	To analyze historical Olympic data (medals.csv, athlete.csv, athlete_events.csv) using data analytics and interactive visualizations to uncover performance trends, medal distributions across nations, and correlations between athlete physical attributes and sports success.
Scope	The project covers Olympic data preprocessing, exploratory data analysis (EDA), statistical aggregations, dynamic KPI calculations, and interactive visual dashboards (bar charts, line graphs, scatter plots) using Python, SQL, and Power BI / Tableau.
Problem Statement	
Description	Historical Olympic data is stored across multiple fragmented CSV files containing over 270,000 raw records with missing values, making it difficult for sports analysts and researchers to gain consolidated insights.
Impact	Without a centralized data pipeline and visual analytics dashboard, sports researchers face slow manual processing, limited dynamic filtering, and difficulty in identifying physical attribute patterns and national dominance trends.
Proposed Solution	
Approach	Ingest raw datasets into a Python/SQL pipeline for automated data

	cleaning and transformation, then import the cleaned data into Tableau / Power BI to build interactive dashboards with multi-filter dynamic exploration.
Key Features	1. **Data Cleaning Pipeline:** Automated missing value handling and duplicate removal. 2. **Dynamic KPI Cards:** Instant tracking of total medals and athlete tallies. 3. **Interactive Visualizations:** Country-wise medal trends, physical attribute analysis (Height/Weight/Age), and sport-specific drill-downs.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Intel Core i5 / AMD Ryzen 5 or higher (Multi-core Processor)	4 CPU Cores @ 2.5GHz or above
Memory	System RAM for dataset ingestion and rendering	8 GB RAM or above
Storage	Disk space for raw datasets, transformed databases, and BI workbooks	256 GB SSD
Software		
Frameworks	Data processing and analytics libraries	Python 3.x, Pandas, NumPy
Libraries	Data visualization and database connectors	Matplotlib, Seaborn, SQLAlchemy, PyMySQL
Development Environment	IDE, Query Editor, and BI Tools	Jupyter Notebook, VS Code, Power BI / Tableau Desktop,

		Git
Data		
Data	Source, size, and format of Olympic datasets	Kaggle Olympic Datasets (athlete_events.csv, medals.csv, athlete.csv), ~48 MB, CSV format